

NHLANHLA MOKOENA

AI / Machine Learning Engineer - Agentic AI, RAG & MLOps

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PROFILE

AI Engineer building production-ready AI systems with LLM APIs, agentic workflows, RAG and end-to-end MLOps. I work with LangChain, LangGraph and structured tool use on the orchestration side, ship systems behind production Python/FastAPI services, and containerise and automate their delivery with Docker and CI/CD. Comfortable owning a system from data pipeline through model training, inference and deployment, and translating unrefined or evolving requirements into working software. Currently applying this to XR simulation and assessment platforms at Nudle, and to systematic evaluation of model behaviour on reasoning, programming and agent tasks at Alignerr.

CORE TECHNICAL SKILLS

Agentic AI & GenAI	LangChain, LangGraph, RAG pipelines, Pinecone VectorDB, tool use, structured outputs, multi-step reasoning, prompt engineering, Claude / Anthropic API, OpenAI API, Groq
ML & Deep Learning	PyTorch, Scikit-learn, XGBoost, transfer learning (ResNet50), fine-tuning, model evaluation, Pandas, NumPy, Matplotlib
MLOps & Delivery	Docker, GitHub Actions CI/CD, end-to-end MLOps pipelines, serverless deployment (HuggingFace Spaces, Render, Vercel)
Backend & Data	Python, FastAPI, PostgreSQL, Redis, SQL, ETL / API integration pipelines, asynchronous task orchestration (asyncio)
Cloud & Web	AWS (IAM, networking, CloudFormation, cost management, ExploreAI Academy coursework; hands-on Terraform provisioning), Azure fundamentals, serverless deployment, Next.js, React, TypeScript, WebXR / React Three Fiber
Ways of working	System design, end-to-end ownership, cross-functional collaboration, working from ambiguous specs, internal tooling for non-technical users

EXPERIENCE

Junior AI Engineer | Nudle May 2026 – Present

- Cut 3D asset turnaround for simulation training from days to hours by building pipelines that convert text or images into GLB assets delivered into XR environments via WebXR.
- Mapped natural-language instructions to avatar actions and animation logic with explainable AI-driven reasoning, improving simulation realism and behavioural consistency.
- Removed the need for engineering support in customising simulations by building systems that generate training scenarios, role-play dialogues and learning tasks from prompts.
- Reduced manual grading workload across learner scenarios through automated assessment and feedback tooling.

AI Trainer (Freelance) | Alignerr Jan 2026 – Present

- Improved production model reliability on reasoning, programming and agent-behaviour tasks by systematically evaluating model outputs against strict quality rubrics.
- Reduced recurring failure modes in agentic and tool-use workflows by authoring structured evaluation feedback used in model training cycles.

Junior AI Software Developer (Contract) | Artintel Jan 2025 – Jul 2025

- Built backend systems for a no-code LLM fine-tuning platform, enabling non-technical users to fine-tune models without engineering support.
- Consolidated training-data handling into a single ETL pipeline by integrating external APIs and databases, cutting manual data-prep steps.

SELECTED PROJECTS

Avatar-3D Pipeline — AI-driven 3D interaction engine | [avatar-pipeline.vercel.app](#)

- Architected an end-to-end “Director & Marionette” pipeline in FastAPI with Groq (Llama-3.3-70b), translating natural language into 14 deterministic 3D skeletal animation states via strict Pydantic JSON validation, structured-output tool invocation under hard schema constraints.
- Built a Next.js 16 / React Three Fiber frontend, resolving WebGL SSR collisions and implementing 0.5s animation crossfading.
- Dockerised the Python backend for immutable deployment on HuggingFace Spaces against a Vercel-hosted frontend.

Orbit-3D Asset Pipeline — Multimodal text/image-to-3D system | [orbit-3d-pipeline.vercel.app](#)

- Engineered a multimodal generation pipeline (Next.js 16, FastAPI, Tripo3D, Groq Llama-4 Vision) producing optimised 3D models from text or images.
- Dockerised a headless Blender processing engine to centre, scale and format AI-generated meshes for WebGL rendering.
- Architected asynchronous task execution with `asyncio.gather` to minimise end-to-end latency.

Real Estate Price Predictor — End-to-end MLOps system | [github.com/Murci20965/real_estate_price_predictor](#)

- Built a deployable end-to-end MLOps system using gradient boosting (XGBoost), achieving $R^2 = 0.9037$ and RMSE = 0.1341 on unseen test data.
- Implemented a CI/CD pipeline with GitHub Actions and Docker, reducing manual deployment time.

Medical Image Classifier — End-to-end MLOps system | [github.com/Murci20965/medical_image_classifier](#)

- Built a chest X-ray Pneumonia/Normal classifier using ResNet50 transfer learning; 82.85% test accuracy with 0.91 precision and 0.96 recall for pneumonia detection.
- Shipped a FastAPI inference service with a Gradio UI, fully Dockerised for reproducible deployment.

EDUCATION & CERTIFICATIONS

ALX/ExploreAI Academy — Data Science

DynamicDNA ICT Academy — System Development

Certifications: Introducing Generative AI with AWS (Udacity) · Microsoft Certified: Azure Fundamentals · Microsoft 365 Certified: Fundamentals · ALX/ExploreAI Certified Data Scientist · Credit Risk Modelling in Python (365 Data Science) · AI Career Essentials (ALX Africa) · Supervised & Unsupervised Learning · Working with OpenAI API · Prompt Engineering with OpenAI API · Data Manipulation with Pandas · Working with Hugging Face